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```
% covariance of gaussian vector
```

```
Sigma1 = eye(2);  
Sigma2 = 1/10*diag([1,10]);  
Sigma3 = rand(2,2); Sigma3 = Sigma3*Sigma3';  
mu = [0;0];
```

```
X = mvnrnd(mu,Sigma1,2000);  
%  
scatter(X(:,1),X(:,2),50, 'b')  
xlim([-4,4]);  
ylim([-4,4]);  
axis off
```



```
X = mvnrnd(mu,Sigma2,2000);  
%  
scatter(X(:,1),X(:,2),50, 'b')  
xlim([-4,4]);  
ylim([-4,4]);  
axis off
```



```
X = mvnrnd(mu,Sigma4,2000); % scatter(X(:,1),X(:,2),50,'.') xlim([-4,4]); ylim([-4,4]); axis off
```

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